SAMARTH

CUET PG 2027

Lecture - 06 | Computer Science

Unit 03 : Digital Fundamental

Topic : Computer Arithmetic

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Recap *of previous lecture*

1. Fixed Point Representation



Topics *to be covered*

1. Computer Arithmetic

Arithmetic operations.
(Addition, Subtraction, multiplication, Division)



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Arithmetic Operations

(1) Unsigned no. → 0 (Positive).

(2) Signed no's (Positive, Negative)

(1) By default a no is always positive.

Ex: 10

By default +10

Ex: n = 3 bits

$= 2^3 - 1$

$= 8 - 1$

$= 7$

(1) Unsigned no's

0	0	0	= 0 ✓
0	0	1	= 1
0	1	0	= 2
0	1	1	= 3
1	0	0	= 4
1	0	1	= 5
1	1	0	= 6
1	1	1	= 7 ✓

Q Direct- Expected

Range of unsigned no

0 to $2^n - 1$

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Sign 0 — Positive no.
Sign 1 — Negative no.

② Signed no

MSB Sign				MSB.			
0	0	0	= +0	1	0	0	= −0
0	0	1	= +1	1	0	1	= −1
0	1	0	= +2	1	1	0	= −2
0	1	1	= (+3)	1	1	1	= (−3)

Range is $-2^{n-1}+1$ to $2^{n-1}-1$

n=3.
$-2^{n-1}+1$
-2^2+1
$-4+1$
(−3)
$2^{n-1}-1$
$2^{3-1}-1$
2^2-1
$4-1$
= (3)

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Addition of 2 unsigned no's

① Binary Half Adder.

A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Ex

$$\begin{array}{r} 1\ 0\ 1\ 0 \quad = 10 \\ +\ 0\ 0\ 1\ 1 \quad = 3 \\ \hline 1\ 1\ 0\ 1 \quad = 13 \end{array}$$

② Octal (0-7).

Ex

$$\begin{array}{r} 6\ 4\ 2 \\ +\ 3\ 2\ 1 \\ \hline 9\ 6\ 3 \end{array}$$

Decimal no ← 9

Convert this to octal.

$(9)_{10} = (11)_8$

Number System Conversion

@ octal

```
  1 1
  1 4 6
+ 1 3 6
-------
  3 0 4
-------
```

$(12)_{10}$ ↓ $(14)_8$

$(8)_{10}$ ↓ $(10)_8$

8 | 12

1 4 ↑

(14)

Signed Nos

Addition → 4 Type of Ops

(1) Greater no is negative

Ex $(-28)_{10} + (15)_{10} = -28 + 15 = (-13)$

$+28 = 0\,1\,1\,1\,0\,0$

$-28 = 1\,0\,0\,1\,0\,0$

$15 = 0\,0\,1\,1\,1\,1$ +

$1\,1\,0\,0\,1\,1 = -13$

2	28	
2	14	0
2	7	0
2	3	1
	1	1

$28 = 0\,1\,1\,1\,0\,0$

8's complement $-28 = 1\,0\,0\,1\,0\,0$

Cross verify:

1's. 11001100

+1

00110101 = +13.

② Both nos negative

Ex $(-28) + (-15) = (-43)$

$28 = 011100$

$15 = 01111$

$-28 = 100100$

$-15 = 110001$

$101011 = -43$

$15 = 01111$

$8's = 10000$

Cross verify.

101011

$8's: 010101 = +43$

(3) Both Positive

$(+28)_{10} + (+15)_{10}$

$= (43)_{10}$

```
      1 1 1
+28 = 0 0 0 1 1 1 0 0
+15 = 0 0 0 0 1 1 1 1   +
      ---------------
      1 0 1 0 1 1 = +43
      ---------------
```

✓

$-15 = 110001$

(4) Smaller no negative.

$+(28)_{10} + (-15)_{10} = +13$

```
-15 = 1 1 0 0 0 1
+28 = 0 1 1 1 0 0   +
      -------------
    1 0 0 1 1 0 1
      -------------
```

Carry bit discard

$0\,0\,1\,1\,0\,1 = +13$

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Subtraction

Rule:- Change the sign of subtrahend then proceed with the rule of addition.

Ex (1) $6 - 10$

$= 6 + (-10)$

Ans.

(2) $-18 - (-15)$

$-18 + 15 = -3$

Ans.

(3) $22 - (-17)$

$22 + 17 = 39$ ⟶

(4) $-22 + 17$

Ex $7 - 3 = 4$

Minuend, Subtrahend

$7 + (-3)$

$$A - B = A + (2\text{'s complement of } B)$$

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Binary Multiplication.

0 X 0 = 0
0 X 1 = 0
1 X 0 = 0
1 X 1 = 1

Ex: 1110 × 101

1 1 1 1 0
0 0 0 0 X
1 1 1 0 X X +
1 0 0 0 1 1 0 ✓

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Binary subtraction

$0 - 0 = 0$

$1 - 0 = 1$

$1 - 1 = 0$

[$0 - 1 = 1$ & 1 borrow

Ex

$$\begin{array}{r} 1\overset{10}{\cancel{1}}\overset{10}{\cancel{0}}\overset{10}{\cancel{0}} \\ - \quad 0011 \\ \hline 1011 \end{array}$$

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Binary Division

000111 = 7

Ex: 110) 101010 (000111

0

-

1 < 110

10

10 < 110

- 0

101

101 < 110

0

1010 > 110

1010

1001 > 110

- 110

0100

- 110

0110

7)72(6

-42

0

2 | 16 .

2 | 8 0

2 | 4 0

2 | 2 0

1 0

7 - 000111

16 = (10000)

16

8)128(

8

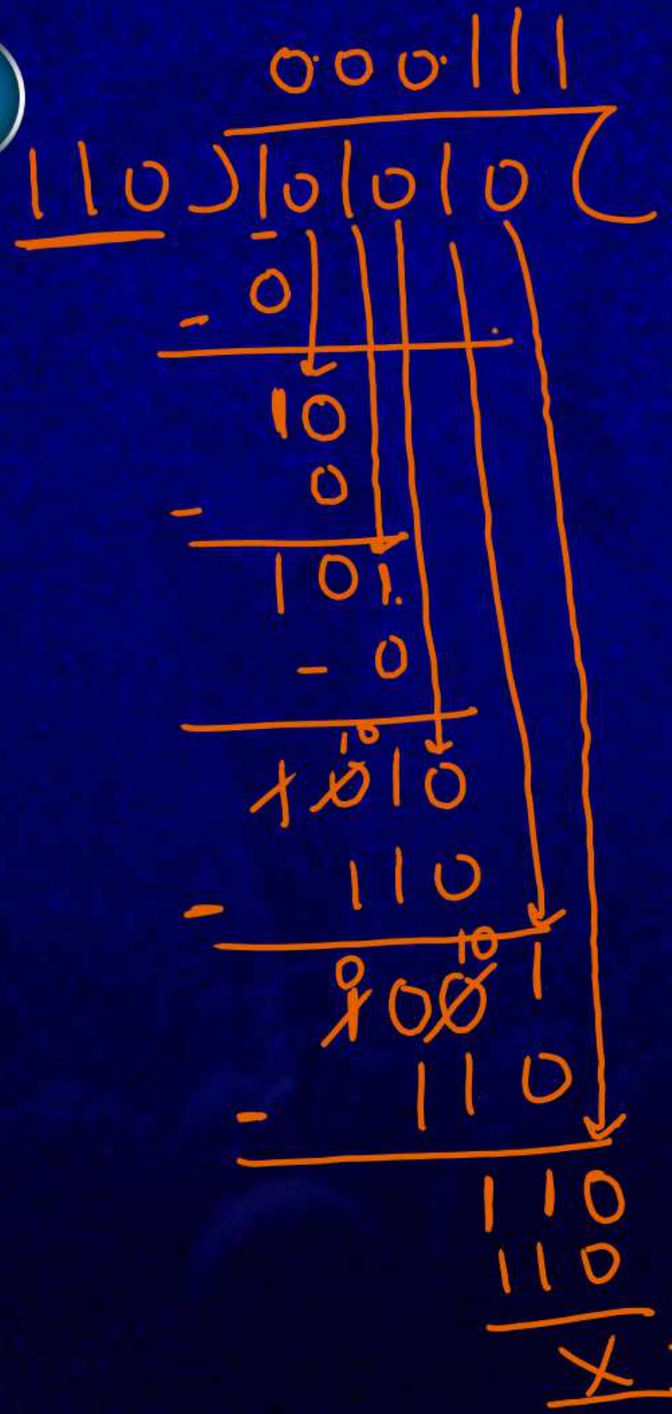
48

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000111
= 7



1 < 110

10 < 110

101 < 110

1010 > 110

1001 > 110

110 = 110

2 Mins Summary

↳ Computer Arithmetic

+ 015

THANK YOU

Point is fixed

12.25

Floating

12.25